

Metasurfaces: High Impedance Surfaces

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Abstract—In this paper, we introduce the high impedance surfaces (HIS), which are metasurfaces that have been found to have abnormal electromagnetic (EM) properties, and thus are of interest in EMC (Electromagnetic Compatibility) applications. We look into the design of a one-patch, one-cell HIS, after which we design a mushroom-type HIS and cavity-type HIS. Finally, we highlight two applications of HIS in shielding box design and antenna design. Through these examples, we demonstrate how HIS can be engineered to reflect EM waves without phase reversal, suppress surface wave propagation, and contribute to optimized EM environments for modern electronic systems.

Index Terms—TEM wave- transverse electromagnetic wave, EMC- electromagnetic compatibility, PMC- perfect magnetic conductor, PEC, perfect electric conductor, AMC- artificial magnetic conductor, grounding plane, patch, EBG- electromagnetic bandgap, HIS- high-impedance surface.

I. INTRODUCTION

With the rapid advancement in PCB technology and chip manufacturing, there arises a greater need for better control of electromagnetic waves. Artificial and periodic structures, metasurfaces, and metamaterials have found application here because of their abnormal EM properties. In this paper, we introduce one such metasurface, high impedance surface (HIS). HIS is a kind of electromagnetic bandgap (EBG) structure, with three important features:

- Can reflect EM waves with no phase reversal, and therefore behaves as an artificial magnetic conductor (AMC).[7]
- Can suppress surface wave (SW) propagation within certain frequency ranges.
- Enables antennas to have good radiation patterns without unwanted ripples.[3]

The HIS consists of patches etched on metal layers of a PCB to form periodic structures (although aperiodically-ordered textures are also possible as in [7]). At a given resonant frequency of these cells, the surface impedance of the HIS becomes very high. This enables it to eliminate surface wave propagation along its surface. Additionally, there are no currents flowing at this resonant frequency, which makes the HIS act like a perfect magnetic conductor, and for this reason, it is sometimes called an artificial magnetic conductor.

II. CLASSICAL DESIGN OF A PATCH-BASED HIS [2]

One of the most common structures is a square patch (although it does not have to be square; it can be any other shape) with or without vias. This structure is easy to simulate and fabricate. Moreover, analytical have been developed to

predict their surface impedance. In this section, we design such a patch. Consider a single patch without any vias, which constitutes a single cell. Perfect Electric Conductor (PEC) conditions are applied to the sides lying in the YZ plane. Perfect Magnetic Conductor (PMC) conditions are applied along the XZ plane. Consequently, a TEM wave is generated with the electric field polarized along the X-axis.

Consider the following parameters:

- Patch length, l_p , is equal to 26.9 mm.
- Gap length, g_l , is equal to 2.4 mm.
- Epoxy substrate, ϵ_r , is equal to 4.3.
- Thickness, h , is equal to 1.58 mm.

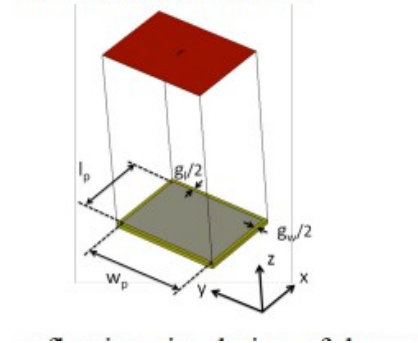


Fig. 1. Unit cell patch HIS [2]

For such a patch, the reflected phase is null at 2.45 GHz (with $\Delta_F = 6\%$). The patch width, W_p , and gap width, G_w , do not influence the surface impedance in this effect. Although this paper does not explore it further, one significant advantage of the HIS is that the patch width, W_p , can be adjusted to fine-tune performance. This is because changing the patch width affects the surface impedance without altering the surface wave reflection or the resonant frequency of the patch.

III. APPLICATIONS OF HIS

A. Uses of HIS

High impedance surfaces (HIS) have several practical uses:

- Used as power ground planes for simultaneous switching noise mitigation in high-speed circuits.
- Employed to enhance the gain of antennas and decrease backward radiation.

A TL (transmission line) model is proposed in [4] for the simple and accurate analysis of the HIS. Although it shows potential benefits, there are still challenges for the HIS to be used in practical engineering.

B. Challenges

One significant challenge of HIS is that cells tend to have a large size, which must be considered given the space constraints in many applications.

IV. TYPES OF HIS

A. Mushroom-Type HIS

Mushroom-type HIS are formed by a grounded wire medium (WM) slab in conjunction with a capacitive grid. The top metal layers of the PCB are etched into many patches, and each patch is connected to the bottom metal ground plane through a via. The gaps between adjacent patches form capacitance C , whereas the vias and ground plane form inductance L . Thus, the unit cell of the HIS can be equivalent to an LC parallel circuit as shown below.

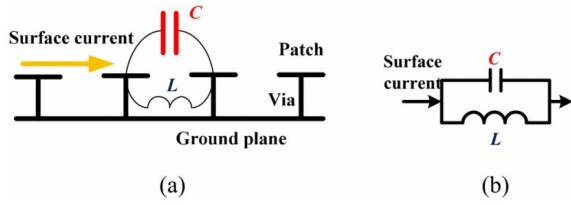


Fig. 2. Diagram of the Mushroom type HIS and its equivalent circuit [1]

The resonant frequency of the LC parallel circuit is given by:

$$\omega_0 = \frac{1}{\sqrt{LC}} \quad (1)$$

For $\omega < \omega_0$, the impedance magnitude of L (X_L) is smaller than that of C (X_C), so a large amount of the current flowing on the HIS surface will go through L :

$$X_C < X_L \quad (2)$$

For $\omega > \omega_0$, the $X_C > X_L$, so the HIS presents a capacitive surface and can support transverse electric (TE) wave propagation:

$$X_C > X_L \quad (3)$$

When $\omega = \omega_0$, the impedance of the HIS approaches infinity:

$$\text{Impedance} \rightarrow \infty \quad (4)$$

This condition stops current flow along the surface and eliminates surface wave propagation along the surface.

B. Cavity-Type HIS

Besides the lumped elements, LC, the patch cavity resonance can also be used to form a HIS. The cavity-type HIS is analogous to the mushroom-type HIS but without the shorting wires. Surface current flows on the patch and the ground plane.

The distance between patches is large enough to avoid parasitic capacitance. The distance between the patch and ground plane is usually larger than the wavelength of interest. For this

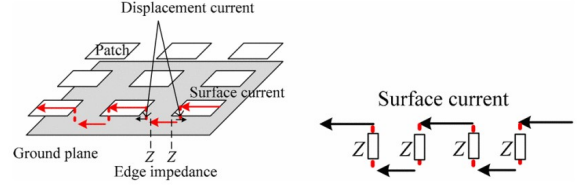


Fig. 3. Diagram of the cavity-type HIS [1]

reason, each patch acts as a perfect electric conductor (PEC) on the top and bottom and a perfect magnetic conductor (PMC) along the edges of the patch. At the resonant frequency, the edge impedance approaches infinity. This greatly suppresses surface wave current and surface wave propagation.

V. APPLICATIONS OF HIS

A. Shielding Box

Metal shielding boxes face several electromagnetic compatibility (EMC) issues:

- Unwanted EM feedback from the output port of the power chip to the input port.
- Changes in total amplifier gain.

A cavity-type HIS is applied to address these problems, as highlighted in [1]. An HIS attached to the upper internal surface of the box acts as a perfect magnetic conductor (PMC) boundary, which suppresses surface wave propagation at the resonant frequencies of the cavity.

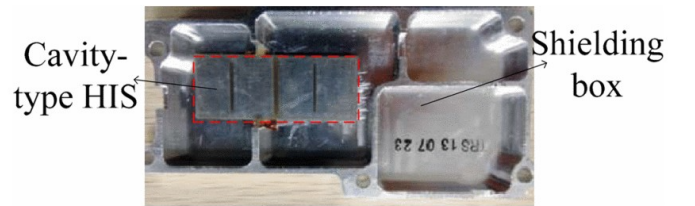


Fig. 4. The fabricated cavity-type HIS, which is attached to the shielding box.[1]

B. Antenna Design

Antennas often suffer from backlobe radiation, as highlighted in [1]. This backlobe radiation has the following effects on antenna design:

- Reduces antenna gain.
- Introduces unwanted coupling between patch antennas and nearby circuits.

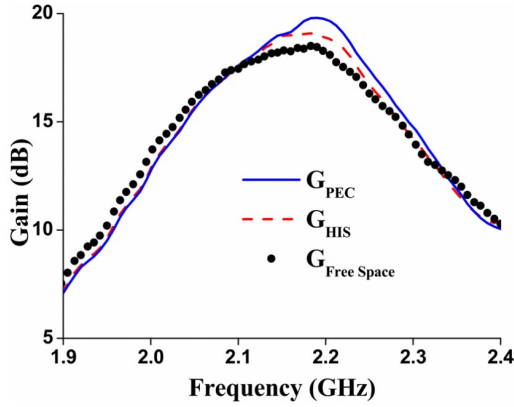


Fig. 5. Measured amplifier gains with the original shielding box GPEC, the HIS attached shielding box GHIS, and without the shielding box Gfree space. [1]

A mushroom-type HIS can be designed to suppress surface wave propagation and improve the performance of antennas operating at Wi-Fi frequencies.

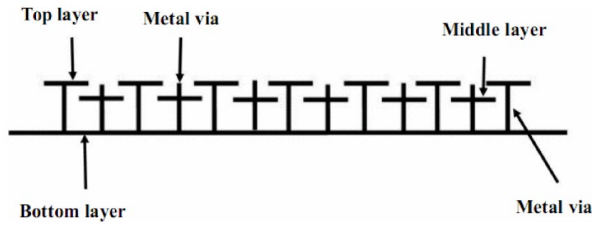


Fig. 6. Side view of 3 layer mushroom-type HIS in antenna design [1]

VI. CONCLUSION

In this paper, we have explored high impedance surfaces (HIS) and their applications in various fields, including electromagnetic compatibility (EMC) and antenna design. We discussed the fundamental principles behind HIS, such as their ability to reflect electromagnetic waves without phase reversal and suppress surface wave propagation.

We examined two primary types of HIS: the mushroom-type and cavity-type HIS. Each type offers distinct advantages and applications, from enhancing antenna performance to addressing EMC challenges in metal shielding boxes. The design considerations and practical implementations of these HIS structures highlight their versatility and effectiveness in modern electronic systems.

To achieve optimal performance with HIS, it is crucial to consider the co-design of the HIS and its surrounding components. This integrated approach ensures that the HIS functions effectively in its intended application.

ACKNOWLEDGMENTS

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